

Paper 9 — Thyroid Synthetic-Lethal Vulnerability Map (Ruppín-style, conceptual)

Date: 2026-05-04 Author: Seungho Cook Status: STRATEGIC DESIGN ONLY. No data download. No analysis. No GPU. No manuscript text. Scope: post-marathon (after 2026-06-13). Anchor on Paper 1 once it is in print. Forbidden in this document: copying ISLE/SELECT prose, claims, figures, or signatures. Concepts are referenced; sentences are not.

I. Executive verdict

GO, conditional on Paper 1 in print.

Paper 9 is a **forward-looking translational paper** that takes the DM1 / RAI-lineage-collapsed state defined in Paper 1 as a **state-of-vulnerability** and asks: "given this state, what genes/drugs kill the tumor selectively, and which compensatory partners must we co-target?" The design is Ruppín-style in spirit (transcriptome-guided synthetic lethality / synthetic rescue, population-level co-occurrence patterns, cross-cohort transferability) but specifically thyroid-tuned and DM1-anchored.

Why this is a real paper and not an extension memo: - Paper 1 defines a state but does not prosecute drug vulnerability. - Existing pan-cancer SL inference engines (ISLE, SELECT, etc.) systematically under-sample thyroid; their inferred pairs are not directly transferable. - DM1-like dedifferentiation (FOXE1/NKX2-1/PAX8/HHEX collapse + STAT3/AP-1/DNMT activation) is an unusual transcriptional state — lineage collapse plus compensatory rewiring — and is precisely the regime where SL theory has the most leverage (lost redundancy → narrow survival corridor).

Risks: lack of thyroid-specific CRISPR depth in DepMap; PRISM/GDSC drug coverage is sparse for thyroid; SL inference is statistically delicate. These are *manageable* but require honest framing (Sec 12).

2. Why Paper 9 exists after Paper 1, not in parallel

PAPER	ASKS	ESTABLISHES
Paper 1	What is DM1?	A reproducible transcriptional state + 8-gene readout
Paper 2	How does Hashimoto-overlap PTC project onto DM1 morphologically/biologically?	DM1 visibility on H&E + immune layer
Paper 3	Why is the dark thyroid cancer ICI-vulnerable?	Immunological vulnerability axis
Paper 4 (backlog)	Korean GD HLA Pan-Asian	independent immunogenetic line
Paper 9	Given DM1, what kills it?	Druggable / wet-lab-testable vulnerability map

Sequencing rationale: - Paper 9 needs the DM1 state to be *citable*, not "in prep". Otherwise reviewers fight the state definition instead of the SL claim. - Paper 1 in print = anchor citation. Paper 9 inherits authority. - Running Paper 9 in parallel would dilute Paper 1's identity ("8-gene readout becomes the basis for a drug map") – better as the *next* paper's headline.

3. Ruppin-style conceptual template (used as scaffolding, not as text)

We borrow concepts and method-class, not language. The conceptual scaffolding is publicly described in the field and may be re-implemented as long as we do not lift sentences, claim independent priority over their algorithms, or imply equivalence to ISLE/SELECT outputs.

Concepts we adopt: 1. **Synthetic lethality (SL) as a population-level statistical signal**. Pairs (A, B) where co-loss is depleted in tumors that survive (because co-loss kills the cell) – operationalized via co-occurrence / co-expression / survival statistics. 2. **Synthetic rescue (SR)**. Partner whose up-regulation rescues fitness when the target is lost; predicts resistance. 3. **Transcriptome as therapy-selection layer**. Move beyond actionable mutations: use gene-expression state to predict drug response, especially in mutation-poor tumors (which DM1 partially is). 4. **Cross-data**

triangulation. SL/SR candidates must show a coherent signal across (i) cell-line dependency, (ii) drug response, (iii) patient outcome – not just one.

Concepts we explicitly *do not* adopt or claim: - We do not claim ISLE-equivalent or SELECT-equivalent inference. - We do not lift their precomputed SL pair lists as ground truth. - We do not benchmark our pipeline against theirs as a horse-race; we treat them as inspiration, not adversary.

If Ruppin-tools are run *for sanity check* (recommended but optional), they are cited as third-party reference, not as our pipeline.

4. Thyroid-specific adaptation

Why thyroid breaks pan-cancer SL out of the box: - Thyroid lines in DepMap are few (≈ 12 named, fewer with full CRISPR depth); pan-cancer SL pairs are dominated by signal from over-represented lineages (breast, colon, lung). - Thyroid mutation density is low; co-mutation-based SL inference is anemic. - Thyroid lineage TFs (PAX8, FOXE1, NKX2-1) are not standard SL nodes in pan-cancer maps.

Adaptations: 1. **State-conditioned SL.** Instead of asking "is gene X SL with Y pan-cancer", ask "is X SL with Y *given a DM1-like transcriptional context*". Define DM1-like state at the cell-line level using the 8-gene compact readout + TF-collapse score, then condition SL inference on that state. 2. **Lineage-imputed pseudobulk.** Augment thin thyroid cell-line panel with thyroid PDX/organoid expression where available, treated as held-out validation, not training. 3. **Cross-tissue borrowing for power.** Where the DM1-like transcriptional state appears in non-thyroid lineages (e.g., dedifferentiated neuroendocrine, lineage-plastic prostate), borrow strength for SL inference but always validate the hit *back* in thyroid context. 4. **TF-collapse as the SL anchor, not driver mutation.** This is the single biggest conceptual departure from canonical Ruppin work, which is mutation-anchored. We are state-anchored.

5. Main hypothesis (single sentence + decomposition)

H0: DM1-like thyroid tumors carry a non-redundant set of synthetic-lethal vulnerabilities arising from lineage collapse (FOXE1/NKX2-1/PAX8/HHEX loss) and compensatory rewiring (STAT3/AP-1/DNMT activation), and these vulnerabilities are predictable from the 8-gene compact readout without requiring driver-mutation profiling.

Decomposition: - **H1 (state)**: DM1-like state is reproducibly callable in CCLE. - **H2 (dependency)**: A subset of CRISPR dependencies in DepMap is significantly enriched in DM1-like lines. - **H3 (compensation)**: A subset of these dependencies has a synthetic-rescue partner whose up-regulation predicts resistance. - **H4 (drug)**: Drug response in PRISM/GDSC/CTRP recapitulates H2 for druggable targets. - **H5 (outcome)**: In TCGA-THCA and GSE76039, high target-pathway expression in DM1-classified tumors stratifies survival – motivating but not proving therapeutic benefit.

6. Candidate target classes

For each: rationale (DM1-link), prior thyroid evidence, drug class, what would falsify, SL-vs-non-SL classification.

6.1 NAMPT / NAD salvage

- **Rationale**: STAT3-high + high transcriptional output → high NAD turnover. NAMPT is the rate-limiter of NAD salvage. Cancers with metabolic plasticity often become NAD-addicted.
- **Drugs**: FK866 (APO866), KPT-9274, OT-82.
- **SL?** Yes, candidate SL with PARP-pathway / OXPHOS partners; SR with NAPRT (NAD de novo arm) – NAPRT-low + NAMPT-high is the classic vulnerability window.
- **Falsifier**: if NAPRT is uniformly high in DM1 lines, vulnerability vanishes.

6.2 STAT₃ / JAK

- **Rationale**: Direct readout of the DM1 inflammatory-rewiring signature.
- **Drugs**: ruxolitinib (JAK1/2), TYK2-selective (deucravacitinib-class), STAT3-selective (TTI-101, napabucasin caveat).
- **SL?** Often *not* SL – STAT3 inhibition alone gives partial response. Real SL targets are likely **upstream cytokine receptors (IL6R, OSMR)** or **parallel TFs (NFκB)** in DM1 context.
- **Falsifier**: STAT3 single-agent inhibitor sensitivity does not separate DM1-like vs. non-DM1 lines.

6.3 DNMT / epigenetic repression

- **Rationale**: DNMT activation in DM1 → hypermethylation of lineage TFs (FOXE1 promoter is a known target). DNMT inhibitors may *reactivate* lineage program.
- **Drugs**: decitabine, guadecitabine; combination with HDACi (vorinostat) or LSD1i.

- **SL?** Conceptually different: this is **lineage-rewiring**, not pure SL. We classify it as a **redifferentiation vulnerability** and frame separately.
- **Falsifier:** DNMTi does not restore FOXE1/PAX8/NKX2-1/HHEX expression in DM1-like cell lines.

6.4 LYN / SRC-family kinases

- **Rationale:** Compensatory tyrosine-kinase signaling in mutation-poor dedifferentiated thyroid; SFK activation is reported in advanced/anaplastic thyroid cancer.
- **Drugs:** dasatinib (broad SFK), bosutinib, saracatinib.
- **SL?** Plausible SL partner with PI3K/AKT or with STAT3.
- **Falsifier:** SFK dependency in DepMap is not enriched in DM1-like lines.

6.5 KCNN4 (KCa3.1)

- **Rationale:** Calcium-activated K⁺ channel; up-regulated in some dedifferentiated solid tumors; reported in immune-tumor crosstalk.
- **Drugs:** senicapoc, TRAM-34 (tool compound).
- **SL?** More likely a *modifier* than a hard SL; included because of CRISPR-screen reproducibility in adjacent contexts.
- **Falsifier:** No DepMap dependency signal at all in DM1-like context.

6.6 DNA damage / PARP

- **Rationale:** Lineage plasticity often correlates with replication stress and BRCA-ness signatures.
- **Drugs:** olaparib, talazoparib, niraparib; ATR inhibitors (ceralasertib).
- **SL?** Yes if replication-stress signature is elevated in DM1; SR partner is BRCA1/2 / HR pathway intact tumors that resist.
- **Falsifier:** No HRD signature / no replication-stress markers (CHEK1, ATR, RPA pathway) elevated in DM1-like tumors.

6.7 Redox / OXPHOS-glycolysis compensation

- **Rationale:** Lineage collapse often forces metabolic reprogramming. DM1 may rely on either OXPHOS (mitochondrial) or compensatory glutamine/glycolysis.
- **Drugs:** IACS-010759 (OXPHOS), CB-839 (glutaminase), 2-DG (research tool).
- **SL?** Complementary metabolic SL pair: GLS1 (glutaminase) ↔ MYC, or OXPHOS ↔ glycolytic capacity (LDHA/HK2).

- **Falsifier:** Metabolic flux signature in DM1 is no different from generic high-proliferation thyroid lines.

6.8 TROP2 — explicit non-SL ADC vulnerability (carved out)

- **Rationale:** TROP2 is a tumor-cell-surface marker, not a synthetic-lethal partner.
 - **Drugs:** sacituzumab govitecan (ADC, payload SN-38).
 - **SL? NO.** This is *expression-level vulnerability* for ADC delivery. We include it in the paper as a **parallel translational opportunity**, explicitly distinguished from the SL-map. Mixing the two categories has been a recurring failure mode in similar papers and we will not.
 - **What we say:** "TROP2-high DM1 tumors are candidates for ADC therapy, by an orthogonal mechanism to the synthetic-lethal targets above."
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7. Data needed later (no download now)

Acquisition is **post-marathon**. List for planning only.

SOURCE	USE	ACQUISITION
DepMap CRISPR (DependencyMap, Broad)	Per-gene dependency in cell lines, conditioned on DM1-like state	Public; figshare / depmap.org
CCLE expression	Define DM1-like cell lines	Public
PRISM Repurposing	Drug-response across ~5,000 compounds, broad cell-line panel	Public
GDSC1/GDSC2	Drug-response, ~700 compounds, deeper IC50 fits	Public
CTRP v2	Redundant drug-response cross- check	Public
TCGA-THCA RNA + mutation + clinical	Patient-cohort validation; outcome stratification	dbGaP/UCSC Xena (already locally available)
GSE76039	Advanced-disease replication; we already use this in Paper 1	Already on disk
Thyroid PDX expression panels (where published)	Held-out validation, not training	Selective public sets
Organoid datasets (where available)	Wet-lab grounding for top hits	Selective public

What we will NOT pull until Paper 1 is in print: - DepMap, PRISM, GDSC, CTRP, organoid sets – all deferred. This document reserves the slot, it does not start the work.

8. Analysis plan (when work begins)

Sequenced, not started. Each step has an explicit GO/NO-GO criterion.

Step A – Define DM1-like state at cell-line level. - Score CCLE lines on the 8-gene compact readout + TF-collapse composite (FOXE1/NKX2-1/PAX8/HHEX). - Calibrate against thyroid lines we already trust as DM1-like (anaplastic, poorly differentiated). - GO if ≥ 6 thyroid lines + ≥ 30 cross-tissue lines hit DM1-like threshold. - NO-GO recovery: relax cross-tissue threshold; re-anchor on TF-collapse only.

Step B – Compute target dependency. - For each gene, contrast CRISPR dependency score in DM1-like vs. non-DM1-like lines. - Multiple-testing correction across the genome. - Stratify by lineage to detect thyroid-specific vs. shared. - GO if any of {NAMPT, NAPRT, STAT3, JAK1/2, DNMT1, DNMT3B, LYN, FYN, SRC, GLS1, ATR} shows DM1-conditioned enrichment.

Step C – Infer SL / SR pairs. - For top dependencies, identify partners whose co-loss / co-up-regulation modulates dependency. - Cross-validate against Ruppian pan-cancer SL pair lists *as a sanity probe only* – not as ground truth. - GO if at least 2 partner-gated SL pairs survive.

Step D – Rank druggable vulnerabilities. - Cross-reference Step B/C hits with PRISM/GDSC drug-target maps. - Score by: (i) effect size in DM1-like lines, (ii) target druggability, (iii) clinical-stage drug availability, (iv) thyroid-cell-line confirmation. - Output: ranked top-10 vulnerability list.

Step E – Validate in patient cohorts. - In TCGA-THCA: stratify by DM1 classifier; test whether high target-pathway score associates with survival, recurrence, RAI-refractoriness flags. - Replicate in GSE76039 (advanced-disease, smaller N). - GO if at least 1 target shows concordant direction across both.

Step F – Stage results into figure plan F1–F6.

9. Figure plan F1–F6

- **F1 – Concept.** Schematic: DM1 state → lost redundancy → narrowed compensatory pathways → druggable nodes. Adapted from Ruppian-style SL/SR concept; visually distinct from any published ISLE/SELECT figure.
- **F2 – DM1-like state in CCLE.** Heatmap of 8-gene + TF-collapse score across lines; thyroid lines flagged; cross-tissue DM1-like lines marked.
- **F3 – Genome-wide CRISPR dependency, DM1 vs. non-DM1.** Volcano. Highlight top 20 hits; color-code by target class (NAD, JAK/STAT, epigenetic, SFK, metabolic, DNA damage).
- **F4 – Top SL/SR pairs.** Ten-row mini-grid: target gene × partner gene, with effect sizes for SL and SR axes.
- **F5 – Drug response (PRISM / GDSC).** IC50 distributions for top druggable targets, DM1-like vs. non-DM1-like. Add NAMPT/NAPRT, JAK, DNMT, SFK panels.
- **F6 – Patient validation (TCGA-THCA + GSE76039).** Survival stratification by DM1 + target-pathway score. Forest plot for hazard ratios.

Supplementary: per-target sensitivity panels; cell-line list with DM1 score; SL/SR statistical tables; sanity-check overlap with Ruppian pan-cancer SL lists (as reference, not benchmark).

10. Wet-lab validation options (not committed; describes the landscape)

Tier 1 – minimum to defend the paper: - siRNA / CRISPR knock-down in 2–3 thyroid cell lines (one DM1-like, one non-DM1-like control) for the top 2 SL targets. - Drug-response curves (IC50 + viability) for matching small-molecule inhibitors.

Tier 2 – strong but optional: - Thyroid organoid panel (if accessible via collaborator) – drug response on top 3 hits. - Combination experiments (target + SR partner inhibitor) to demonstrate the SR rescue logic.

Tier 3 – beyond first paper, for follow-up: - PDX in vivo for top 1 hit. - Patient-derived organoid bank cross-screen.

Decision: paper is *defendable* with Tier 1; Tier 2 is a stretch for first round; Tier 3 belongs to a follow-up.

Cost / collaborator question: thyroid wet-lab access. Open. If no wet-lab partner, paper still publishable as **computational + patient-cohort** vulnerability map, framed as hypothesis-generating, in a translational journal (JCI Insight, Mol Cancer Therap, NPJ Precision Oncology) rather than a flagship outlet.

11. Business / IP angle

- **Composition-of-matter:** not patentable for repurposed approved drugs (NAMPT inhibitors in trials, ruxolitinib, decitabine, dasatinib, olaparib, IACS-010759, CB-839 – all third-party).
- **Method-of-use / method-of-treatment:** potentially patentable: "method of selecting thyroid cancer patients for [target-class] therapy by measuring the 8-gene DM1 readout + TF-collapse score." This is the actionable IP slot.
- **Companion diagnostic:** the 8-gene readout itself is the most defensible asset; CDx claims tied to specific drug classes are filable separately.

- **Trial design:** umbrella / basket trial design where the DM1 score routes patients to one of 3–4 target arms (NAD, JAK/STAT, epigenetic, ADC). Pitchable to industry partners after Paper 9 publishes.
 - **Defensive citation:** cite Ruppín-group methods as the conceptual prior art *generously* to avoid any infringement appearance and to position our work as thyroid-specific, state-anchored, clinically translatable.
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12. What not to claim

Hard rules. Violating any of these wrecks reviewer trust.

1. Do **not** claim direct equivalence with ISLE/SELECT outputs. We are conceptually inspired, not replicating.
 2. Do **not** claim clinical efficacy. Patient-cohort survival stratification is hypothesis-generating, not proof of therapeutic benefit.
 3. Do **not** claim DM1-readout replaces driver-mutation profiling. Frame as **complementary**, especially in mutation-poor / RAI-refractory settings.
 4. Do **not** mix TROP2-ADC vulnerability with synthetic lethality. They are different mechanistic categories. Carved out in Sec 6.8.
 5. Do **not** over-extend to Hashimoto-overlap or Graves' subtypes. That is Paper 2 / Paper 4 territory; Paper 9 is specifically the **DM1 dark-matter axis**.
 6. Do **not** use the H&E-DM1 angle. NO-GO per existing standing decision. Stay in molecular / drug-response / outcome space.
 7. Do **not** claim wet-lab validation we have not done. If Tier 1 is missing, label the paper "computational + cohort-validated, wet-lab pending" explicitly in the abstract.
 8. Do **not** lift sentences, signature lists, or precomputed pair tables from Ruppín-group papers. Concept yes, text/data no.
 9. Do **not** start any of this before Paper 1 is in print and the marathon is closed (2026-06-13 floor).
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13. Why this is Paper 9, not Paper 1 / 2 / 3

QUESTION	ANSWER
Could this be folded into Paper 1?	No. Paper 1 establishes biology; folding in SL inflates scope and invites reviewer mutiny on the state definition.
Could this be Paper 2?	No. Paper 2 is the Hashimoto-overlap PTC paper (HT-isolated per 2026-05-04 Yu decision). Different cohort, different question.
Could this be Paper 3?	No. Paper 3 is the ICI-vulnerability dark thyroid cancer paper (FROZEN bundle). Synthetic lethality is a different therapeutic axis; mixing them dilutes both.
Why number 9 specifically?	Reserves a slot in the long-range pipeline. Papers 5–8 are reserved for [paper-numbering doc owner-defined slots]; Paper 9 is the first translational follow-up to the DM1 axis. The number is conventional, not literal – reassign if numbering changes.
Why not just publish target list as a follow-up letter?	Letter-format does not support the figure-and-cohort weight of F1–F6 + supplements. Full article is the right vehicle.

14. Twelve-week future execution plan (post-marathon)

Floor: cannot start before manuscript marathon closes (2026-06-13). Earliest practical start: 2026-06-16. Twelve-week sprint plan, each week is a checkpoint, not a deadline.

WK	PHASE	OUTPUT
1	DepMap + CCLE pull, integrity checks	Local mirror, version-pinned
2	Define DM1-like state in CCLE; score thyroid lines	DM1 score table, cell-line ranking
3	Genome-wide DM1-conditioned dependency analysis	Volcano (F3 draft)
4	SL / SR pair inference	Pair table, sanity cross-check vs. Ruppin lists
5	PRISM / GDSC pull and integration	Drug-response matrix
6	Drug-response stratification by DM1; F5 draft	Per-target IC50 panels
7	TCGA-THCA + GSE76039 patient validation	Survival forest plots (F6 draft)
8	Cross-checks: lineage, batch, confounders	Sensitivity tables
9	Figure assembly F1–F6, supplements	Figure pack v1
10	Draft writing (intro, results, discussion)	Manuscript v0
11	Internal review with Yu / collaborator(s); revise	Manuscript v1
12	Submission package: cover letter, COI, key resources	Submitted

Wet-lab Tier 1 is a parallel track running weeks 4–12 if a collaborator is available; otherwise the paper is submitted as computational + cohort-validated and Tier 1 is the response-to-reviewer plan.

Stop conditions: - If Step A (DM1-like state in CCLE) fails: pivot to bulk pseudobulk validation only and reframe scope. Do not fabricate state membership. - If Step C (SL pairs) yields zero hits: paper still publishable as a *negative* result + ranked dependency list, but in a lower-tier venue. Decide explicitly at week 4 checkpoint. - If wet-lab Tier 1 fails for top hits: replace top hits with rank-2/3 vulnerabilities, do not bury the failure.

End of strategic plan

This document is design only. No data has been pulled, no analysis run, no GPU launched, no manuscript text written. Paper 1, 2, 3, 4 not touched. Voice-protected sections not authored.

Next action gates (in order): 1. Paper 1 in print. 2. Marathon closed ($\geq$ 2026-06-13). 3. Yu sign-off on Paper 9 slot. 4. Then and only then: Step A.